A SELF-STUDY REFERENCE DECK

Laplace Transforms

From the definition to the tautochrone — a single-volume reference with worked examples, transform tables, and historical context.

Personal study reference | No external attribution

Contents — Six Parts, Fifty-Five Slides

PART I

Foundations

Why Laplace? The two-domain picture. The definition.
Existence and uniqueness. Linearity. The power-series analogy.

Why Laplace? Two Domains and a Black Box

The Pain Point

Classical ODE methods — undetermined coefficients, variation of parameters — require continuity of the forcing function $f(t)$. Many physical inputs are discontinuous or impulsive: switch flips, hammer blows, voltage spikes, point loads. Stitching piecewise solutions by hand is tedious.

The Laplace Strategy

Transform the IVP from the *time domain* t to the *frequency domain* s. The derivative d/dt becomes multiplication by s. Solve the resulting algebraic equation for $Y(s)$. Invert the transform to recover $y(t)$.

Time domain

$f(t), \;\; y(t)$

differential equation

$$\text{Time domain} \;\underset{\mathcal{L}^{-1}\{\cdot\}}{\overset{\mathcal{L}\{\cdot\}}{\rightleftarrows}}\; \text{Frequency domain}$$

Frequency domain

$F(s), \;\; Y(s)$

algebraic equation

Why Laplace? Two Domains and a Black Box (cont.)

The Takeaway

A differential equation becomes an algebraic equation — initial conditions are absorbed automatically, and discontinuous forcing is handled natively.

The Laplace Transform — Definition and Notation

Master Formula

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^{\infty} e^{-st} f(t)\, dt$$

Notation Conventions

- Lowercase f, g, y for time-domain functions
- Uppercase F, G, Y for their Laplace transforms
- s is generally complex: $s = \sigma + i\omega$
- Lower limit 0^- captures impulses $\delta(t)$ at the origin

The 0^- Subtlety

When the integrand includes $\delta(t)$, the lower limit must be 0^- so the impulse is captured. Without it, $\mathcal{L}\{\delta(t)\} = 1$ would fail.

Domain of s

The integral converges absolutely for s large enough: $\mathrm{Re}(s) > \alpha$, where α is the exponential-order bound on f

The Starter Transforms

Six Transform Pairs to Know

$f(t)$	$F(s) = \mathcal{L}\{f(t)\}$	**Valid for**
1	$\frac{1}{s}$	$s > 0$
t	$\frac{1}{s^2}$	$s > 0$
$t^n, \;\; n \in \mathbb{N}$	$\frac{n!}{s^{n+1}}$	$s > 0$
e^{at}	$\frac{1}{s-a}$	$s > a$
$\sin(at)$	$\frac{a}{s^2+a^2}$	$s > 0$
$\cos(at)$	$\frac{s}{s^2+a^2}$	$s > 0$

The Starter Transforms (cont.)

Deriving $\mathcal{L}\{1\}$

$$\mathcal{L}\{1\} = \int_0^\infty e^{-st}\,dt = \left[-\frac{1}{s}e^{-st}\right]_0^\infty = \frac{1}{s}$$

For $s > 0$, the boundary term at ∞ vanishes.

Deriving $\mathcal{L}\{e^{at}\}$

$$\mathcal{L}\{e^{at}\} = \int_0^\infty e^{-(s-a)t}\,dt = \frac{1}{s-a}$$

For $s > a$. Same pattern as $\mathcal{L}\{1\}$ with $s \to s - a$.

The Euler Trick — $\sin$ and $\cos$ from e^{ibt}

Worked Example

Step 1. From the table, $\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$ for $s > a$. Substitute $a = ib$ (where b is real):

$$\mathcal{L}\{e^{ibt}\} = \frac{1}{s - ib}$$

Step 2. Multiply numerator and denominator by the conjugate $s + ib$:

$$\mathcal{L}\{e^{ibt}\} = \frac{s + ib}{s^2 + b^2}$$

Step 3. Apply Euler's formula $e^{ibt} = \cos(bt) + i\sin(bt)$ and use linearity:

$$\mathcal{L}\{\cos(bt) + i\sin(bt)\} = \frac{s}{s^2 + b^2} + i\,\frac{b}{s^2 + b^2}$$

Step 4. Equate real and imaginary parts:

$$\mathcal{L}\{\cos(bt)\} = \frac{s}{s^2 + b^2}, \qquad \mathcal{L}\{\sin(bt)\} = \frac{b}{s^2 + b^2}$$

The Euler Trick — $\sin$ and $\cos$ from e^{ibt} (cont.)

Key Takeaway

Complex s is not a complication — it is a unification. Sines and cosines are projections of complex exponentials, so their transforms follow immediately from $\mathcal{L}\{e^{at}\}$.

When Does $\mathcal{L}\{f\}$ Exist?

Two Sufficient Conditions

(a) f is *piecewise continuous* on every finite $[0, b]$: finitely many jumps, one-sided limits exist.

(b) f is of *exponential order*: $\exists\, M, \alpha, t_0$ such that $|f(t)| \le Me^{\alpha t}$ for $t > t_0$.

Existence Theorem

If f is piecewise continuous on every $[0, b]$ and of exponential order $e^{\alpha t}$, then $\mathcal{L}\{f\}$ exists for $s > \alpha$.

Proof Sketch (Comparison Test)

Step 1. Split: $\int_0^\infty = \int_0^{t_0} + \int_{t_0}^\infty$. The first piece exists by piecewise continuity.

Step 2. For the tail, use exponential order:

$$|e^{-st} f(t)| \le Me^{-(s-\alpha)t}$$

Step 3. The dominating integral converges for $s > \alpha$:

$$\int_{t_0}^\infty Me^{-(s-\alpha)t}\, dt = \frac{M}{s-\alpha}\, e^{-(s-\alpha)t_0}$$

Step 4. Apply the comparison test $\Rightarrow$ $\mathcal{L}\{f\}$ converges absolutely.

Sufficient, not necessary

$f(t) = t^{-1/2}$ is not piecewise continuous at $t = 0$, yet $\mathcal{L}\{t^{-1/2}\} = \sqrt{\pi/s}$ exists via $t = u^2$.

When Is $\mathcal{L}^{-1}\{F\}$ Unique?

Uniqueness Theorem

If f and g are *continuous* on $[0, \infty)$ and of exponential order, and $\mathcal{L}\{f\} = \mathcal{L}\{g\}$ for all $s > C$, then $f(t) = g(t)$ for all $t \geq 0$.

Why we require continuity

The Laplace integral cannot distinguish functions that differ only at finitely many points. Uniqueness holds only for *continuous* inverses — which is fine, since solutions of ODEs are continuous.

Counterexample

Define

$$g(t) = \begin{cases} 1, & 0 < t < 3 \\ 2, & t = 3 \\ 1, & t > 3 \end{cases}$$

Then $\mathcal{L}\{g\} = \frac{1}{s}$ for $s > 0$, which equals $\mathcal{L}\{1\}$. But clearly $g(t) \neq 1$ at $t = 3$.

The *continuous* inverse of $1/s$ is $f(t) = 1$.

Heaviside at zero

$u(0) = 0$, $u(0) = \frac{1}{2}$, and $u(0) = 1$ all give the same transform $1/s$. The value at the discontinuity is invisible.

Laplace Transforms as Continuous Power Series

The Analogy

A power series: $\sum_{n=0}^{\infty} a(n)\, x^n$.

Its natural continuous analog: $\int_0^{\infty} a(t)\, x^t \, dt$.

Substitute $x = e^{-p}$:

$$\int_0^{\infty} a(t)\, e^{-pt} \, dt = \mathcal{L}\{a(t)\}$$

The analogy suggests that Laplace transforms occupy in continuous analysis a role parallel to that of power series in discrete analysis.

Why the Analogy Matters

1. **Convergence theory.** Power series converge inside a radius; Laplace transforms converge in a half-plane $\mathrm{Re}(s) > \alpha$. The half-plane is the continuous analog of the radius.
2. **Operations.** Term-by-term differentiation of a power series corresponds to differentiation under the integral sign of a Laplace transform.
3. **Inversion.** A power series is recovered from its coefficients; a Laplace transform is recovered via the Bromwich inversion integral.

Key Takeaway

The Laplace transform is the natural continuous counterpart of the power series — the most useful discrete representation in analysis.

Linearity and the Product Warning

Linearity Theorem

If $\mathcal{L}\{f\}$ and $\mathcal{L}\{g\}$ exist, then for any constants α, β:

$$\mathcal{L}\{\alpha f + \beta g\} = \alpha\,\mathcal{L}\{f\} + \beta\,\mathcal{L}\{g\}$$

The inverse $\mathcal{L}^{-1}$ inherits linearity.

Combining Transforms

$$\mathcal{L}\{2 + 5t + 9e^{-2t}\} = \frac{2}{s} + \frac{5}{s^2} + \frac{9}{s+2}$$

No integration required — pure table lookup plus linearity.

CRITICAL: $\mathcal{L}\{f \cdot g\} \neq \mathcal{L}\{f\} \cdot \mathcal{L}\{g\}$

A product in the time domain does *not* transform into a product in the frequency domain. The correct product rule is the *convolution theorem*: $\mathcal{L}\{f * g\} = \mathcal{L}\{f\} \cdot \mathcal{L}\{g\}$, where $*$ denotes convolution.

Why the Warning Matters

Wrong: $\mathcal{L}\{t \sin t\} \neq \frac{1}{s^2} \cdot \frac{1}{s^2+1}$.
Right (use $\mathcal{L}\{tf(t)\} = -F'(s)$):

$$\mathcal{L}\{t \sin t\} = -\frac{d}{ds}\left[\frac{1}{s^2+1}\right] = \frac{2s}{(s^2+1)^2}$$

Linearity and the Product Warning (cont.)

Key Takeaway

Linearity is a friend; naive multiplication is an enemy. The transform converts sums into sums — but converts *convolutions* into products.

PART II
The Transform Toolkit

Shifting properties. The Heaviside function. The Dirac delta.
Transforms of derivatives and integrals. The full operational calculus.

First Shifting Property (Translation in s)

The Rule

$$\mathcal{L}\{e^{at}f(t)\} = F(s-a)$$

where $F(s) = \mathcal{L}\{f(t)\}$. Inversely, $\mathcal{L}^{-1}\{F(s-a)\} = e^{at}f(t)$.

How to Use It

A denominator like $(s-a)^2 + b^2$ is recognized as s^2+b^2 shifted by a. The inverse is $e^{at}\sin(bt)$. Same trick for $(s-a)^{n+1} \to e^{at}t^n/n!$.

$\mathcal{L}\{e^{at}\sin(bt)\}$

From the table, $\mathcal{L}\{\sin(bt)\} = \frac{b}{s^2+b^2}$. By the first shifting property:

$$\mathcal{L}\{e^{at}\sin(bt)\} = \frac{b}{(s-a)^2+b^2}, \quad s > a$$

$\mathcal{L}^{-1}\left\{\frac{1}{s^2+4s+8}\right\}$

Complete the square: $s^2+4s+8 = (s+2)^2+4$. So

$$\frac{1}{s^2+4s+8} = \frac{1}{(s+2)^2+2^2}$$

Recognize $\frac{b}{(s-a)^2+b^2}$ with $b=2, a=-2$, missing a factor of 2:

$$\mathcal{L}^{-1}\left\{\frac{1}{(s+2)^2+4}\right\} = \frac{1}{2}e^{-2t}\sin(2t)$$

First Shifting Property (Translation in s) (cont.)

Key Takeaway

Whenever a denominator has the form $(s-a)^2+b^2$ or $(s-a)^{n+1}$, the first shifting property gives the inverse immediately — no partial fractions needed.

Second Shifting Property (Translation in t)

The Rule

$$\mathcal{L}\{u_a(t)\, f(t-a)\} = e^{-as}\, F(s)$$

where $u_a(t) = u(t-a)$ is the Heaviside step at $t = a$, and $F(s) = \mathcal{L}\{f(t)\}$.

Inversely: $\mathcal{L}^{-1}\{e^{-as}F(s)\} = u_a(t)\, f(t-a)$.

How to Use It

If e^{-as} multiplies a recognizable transform, the inverse is the un-shifted inverse *delayed by a* and *zeroed before* $t = a$.

$\mathcal{L}\{u_a(t)\}$

Take $f(t) = 1$. Then $F(s) = 1/s$, so:

$$\mathcal{L}\{u_a(t)\} = \frac{e^{-as}}{s}, \quad s > 0$$

$\mathcal{L}^{-1}\left\{e^{-4s}\left(\frac{2}{s^2} + \frac{5}{s}\right)\right\}$

Recognize $F(s) = 2/s^2 + 5/s$. Its inverse is $f(t) = 2t + 5$. Apply the second shifting property with $a = 4$:

$$\mathcal{L}^{-1}\{e^{-4s}F(s)\} = u_4(t)\, f(t-4) = u_4(t)\,[2(t-4)+5] = u_4(t)\,(2t-3)$$

This is 0 for $t < 4$ and $2t - 3$ for $t > 4$.

Second Shifting Property (Translation in t) (cont.)

Key Takeaway

e^{-as} in the s-domain is a *time delay a* in the t-domain. This is the foundation for switched circuits, delayed forcing, and piecewise inputs.

The Unit Step Function as a Building Block

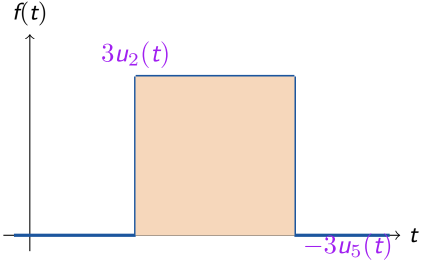
Definition

$$u_a(t) = u(t-a) = \begin{cases} 0, & t < a \\ 1, & t \geq a \end{cases}$$

A jump discontinuity of size $+1$ at $t = a$. The value at the discontinuity is invisible to $\mathcal{L}$.

Three Uses

(a) Switched inputs: $V_0\, u_a(t)$ is a constant voltage switched on at $t = a$.
(b) Box functions: $M[u_a(t) - u_b(t)]$ is M on $[a, b]$ and 0 elsewhere.
(c) Piecewise functions: any piecewise f is $\sum_k u_{a_k}(t)\, f_k(t - a_k)$.



The Unit Step Function as a Building Block (cont.)

A Box Function

$f(t) = 3$ on $[2, 5]$ and 0 elsewhere: $f(t) = 3u_2(t) - 3u_5(t)$.

$$\mathcal{L}\{f\} = \frac{3(e^{-2s} - e^{-5s})}{s}$$

Encoding Piecewise Functions with Heaviside

Single Step

$f(t) = 5$ for $t < 2$, 0 for $t > 2$.
Express as $5(1 - u_2(t))$:

$$\mathcal{L}\{f\} = \frac{5(1 - e^{-2s})}{s}$$

Box Pulse

$f(t) = 0$ for $t < 2$, 3 for $2 < t < 5$, 0 for $t > 5$.
Express as $3u_2(t) - 3u_5(t)$:

$$\mathcal{L}\{f\} = \frac{3(e^{-2s} - e^{-5s})}{s}$$

Triangular Pulse

$f(t) = t$ on $[0, 1]$, $-t + 2$ on $[1, 2]$, 0 elsewhere.
Write $f(t) = t - 2(t - 1)u_1(t) + (t - 2)u_2(t)$:

$$\mathcal{L}\{f\} = \frac{1 - 2e^{-s} + e^{-2s}}{s^2}$$

Pattern

Each piecewise function is a linear combination of $u_a(t)$ terms. The second shifting property then gives each term's transform as e^{-as} times the transform of the shifted function.

The Dirac Delta Function

Definition (as a Limit)

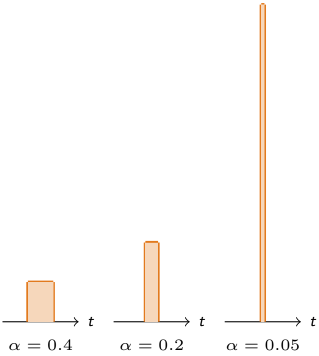
$$\delta(t - t_0) = \lim_{\alpha \to 0} \frac{1}{2\alpha} \left[u_{t_0-\alpha}(t) - u_{t_0+\alpha}(t) \right]$$

A rectangular pulse of width 2α and height $1/(2\alpha)$ — area is always 1.

Defining Properties

$$\delta(t - t_0) = 0 \text{ for } t \neq t_0, \qquad \int_{-\infty}^{\infty} \delta(t - t_0)\, dt = 1$$

Informally: $\delta(0) = \infty$, zero elsewhere, total integral 1. Strictly: δ is a *generalized function* (distribution).



$\delta(t) = \lim_{\alpha \to 0}$ of unit-area pulses

Physical Interpretation

$\delta(t - t_0)$ models an idealized impulse: hammer blow, lightning strike, voltage spike, point load.

The Dirac Delta Function (cont.)

Common Pitfall

$\delta(t)$ is *not* a function in the classical sense. The rigorous foundation is the theory of distributions (Schwartz, 1940s).

The Sifting Property and $\mathcal{L}\{\delta\}$

Sifting Property

For any bounded continuous g:

$$\int_{-\infty}^{\infty} \delta(t - t_0)\, g(t)\, dt = g(t_0)$$

The delta function "sifts out" the value of g at t_0.

δ is the Identity for Convolution

$$(f * \delta)(t) = \int_0^t f(\tau)\, \delta(t - \tau)\, d\tau = f(t)$$

So δ is the multiplicative identity for convolution, even though 1 is *not* (recall $f * 1 \neq f$).

Deriving $\mathcal{L}\{\delta(t - t_0)\}$

Apply the sifting property with $g(t) = e^{-st}$:

$$\mathcal{L}\{\delta(t - t_0)\} = \int_{0^-}^{\infty} e^{-st}\, \delta(t - t_0)\, dt = e^{-st_0}$$

Special case $t_0 = 0$: $\mathcal{L}\{\delta(t)\} = 1$.

Impulse Forcing

Solve $y'' + 2y' + 26y = \delta_4(t)$, $y(0) = 1, y'(0) = 0$ (damped oscillator struck at $t = 4$).
Transform: $s^2 Y - s + 2(sY - 2) + 26Y = e^{-4s}$. Solve:

$$Y(s) = \frac{s + 2}{s^2 + 2s + 26} + \frac{e^{-4s}}{s^2 + 2s + 26}$$

Invert (using $(s + 1)^2 + 25$):
$y(t) = e^{-t}\cos(5t) + \frac{1}{5}e^{-t}\sin(5t) + u_4(t) \cdot \frac{1}{5}e^{-(t-4)}\sin(5(t - 4))$

Transforms of Derivatives — The ODE Engine

The Three Rules

$$\mathcal{L}\{f'(t)\} = sF(s) - f(0)$$
$$\mathcal{L}\{f''(t)\} = s^2F(s) - sf(0) - f'(0)$$
$$\mathcal{L}\{f^{(n)}(t)\} = s^nF(s) - s^{n-1}f(0) - \cdots - f^{(n-1)}(0)$$

Proof Sketch (Integration by Parts)

$$\mathcal{L}\{f'\} = \int_0^\infty e^{-st}f'(t)\,dt$$
$$= \left[e^{-st}f(t)\right]_0^\infty + s\int_0^\infty e^{-st}f(t)\,dt$$
$$= -f(0) + sF(s)$$

The boundary term at ∞ vanishes by exponential order.

Worked Example — Why This Matters

Solve $y'' + 7y' + 10y = 0$, $y(0) = 1, y'(0) = 1$.

Step 1. Transform:

$$[s^2Y - s - 1] + 7[sY - 1] + 10Y = 0$$

Step 2. Solve algebraically:

$$Y(s) = \frac{s+8}{s^2+7s+10} = \frac{2}{s+2} - \frac{1}{s+5}$$

Step 3. Invert: $y(t) = 2e^{-2t} - e^{-5t}$.

The initial conditions are baked in — no separate solve for c_1, c_2.

Key Takeaway

Differentiation in t becomes multiplication by s. A linear ODE with constant coefficients becomes a purely algebraic equation for $Y(s)$.

Transforms of Derivatives — The ODE Engine (cont.)

The 0^- Convention

When impulses $\delta(t)$ are present, write $f(0^-)$ instead of $f(0)$ — the *pre-initial condition*.

Transforms of Integrals — and Integral Equations

The Integral Rule

$$\mathcal{L}\left\{\int_0^t f(\tau)\,d\tau\right\} = \frac{F(s)}{s}$$

Proof. Let $g(t) = \int_0^t f(\tau)\,d\tau$. Then $g'(t) = f(t)$ and $g(0) = 0$. Apply the derivative rule: $\mathcal{L}\{g'\} = sG(s) - g(0) = sG(s)$. But $\mathcal{L}\{g'\} = F(s)$. So $G(s) = F(s)/s$.

Inverting $F(s)/s$

$$\mathcal{L}^{-1}\left\{\frac{F(s)}{s}\right\} = \int_0^t f(\tau)\,d\tau$$

A $1/s$ factor represents a time-domain integration.

A Volterra Integral Equation

Solve $x(t) - t = \int_0^t x(\tau)\,d\tau$.
Transform both sides (using the integral rule on the RHS):

$$X(s) - \frac{1}{s^2} = \frac{X(s)}{s}$$

Solve algebraically: $X(s)\left(1 - \frac{1}{s}\right) = \frac{1}{s^2} \Rightarrow X(s) = \frac{1}{s(s-1)} = \frac{1}{s-1} - \frac{1}{s}$
Invert: $x(t) = e^t - 1$.

A differential equation and an integral equation become the same algebraic equation in the s-domain.

Key Takeaway

Integration in t becomes division by s. Integral equations become algebraic equations just as cleanly as differential equations.

Differentiation and Integration of Transforms

Two More Rules

Differentiation in s:

$$\mathcal{L}\{t\,f(t)\} = -F'(s)$$

$$\mathcal{L}\{t^n f(t)\} = (-1)^n F^{(n)}(s)$$

Integration in s:

$$\mathcal{L}\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(u)\,du$$

$\mathcal{L}\{t\sin(at)\}$

From $\mathcal{L}\{\sin(at)\} = a/(s^2 + a^2)$: $\mathcal{L}\{t\sin(at)\} = -\frac{d}{ds}\left[\frac{a}{s^2+a^2}\right] = \frac{2as}{(s^2+a^2)^2}$
No integration — pure differentiation.

Showcase — Bessel $J_0(x)$ via Laplace

Bessel's equation of order zero: $xy'' + y' + xy = 0$, $y(0) = 1$.
Transform using $\mathcal{L}\{xy''\} = -\frac{d}{ds}[s^2 Y - sy(0)]$ and $\mathcal{L}\{xy\} = -Y'(s)$: $-\frac{d}{ds}[s^2 Y - s] + (sY - 1) - Y' = 0$

$$\implies (s^2 + 1)\,Y' = -sY$$

Separate and integrate: $Y(s) = \frac{c}{\sqrt{s^2+1}}$.
Setting $y(0) = 1$ gives $c = 1$: $\mathcal{L}\{J_0\} = \frac{1}{\sqrt{s^2+1}}$.
Expand by the binomial series and invert term-by-term:

$$J_0(x) = \sum^{\infty} \frac{(-1)^n}{(n!)^2}\left(\frac{x}{2}\right)^{2n}$$

Differentiation and Integration of Transforms (cont.)

Key Takeaway

The rules $\mathcal{L}\{t^n f\} = (-1)^n F^{(n)}$ and $\mathcal{L}\{f/t\} = \int_s^\infty F$ extend the transform's reach to functions multiplied or divided by powers of t — including Bessel functions and many special functions of mathematical physics.

PART III

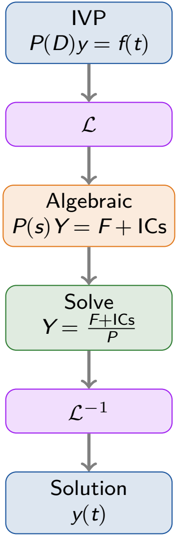
Solving ODEs with Laplace

The five-step method. Worked examples. Discontinuous and impulse forcing. Linear systems. The mass-spring-dashpot application.

The Five-Step Laplace Method

The Five Steps

Step 1 — Transform. Take $\mathcal{L}$ of both sides. Each $y^{(k)}$ becomes $s^k Y(s) - s^{k-1}y(0) - \cdots - y^{(k-1)}(0)$.
Step 2 — Substitute ICs. Plug in the initial conditions. They vanish from the algebraic equation.
Step 3 — Solve algebraically. Collect $Y(s)$ terms and divide.
Step 4 — Decompose $Y(s)$. Use partial fractions, completing the square, shifting properties, or convolution to break $Y(s)$ into table-recognizable pieces.
Step 5 — Invert. Look up each piece in the transform table to recover $y(t)$.



The Five-Step Laplace Method (cont.)

Why This Beats Classical Methods

- Initial conditions are absorbed automatically.
- Discontinuous and impulsive forcing are handled natively.
- The s-domain form exposes the structure of the solution.

Homogeneous ODEs — Three Examples

Distinct Real Roots

Problem. $y'' - 5y' + 6y = 0$, $y(0) = 1, y'(0) = 3$.
Transform: $(s^2 - 5s + 6)Y = s - 2$.
$$Y = \frac{s-2}{(s-2)(s-3)} = \frac{1}{s-3}$$
Answer: $y(t) = e^{3t}$.

Repeated Roots

Problem. $y'' + 2y' + y = 0$, $y(0) = 1, y'(0) = 0$.
Transform: $(s+1)^2 Y = s + 2$.
$Y = \frac{s+2}{(s+1)^2} = \frac{1}{s+1} + \frac{1}{(s+1)^2}$
Answer: $y(t) = (1+t)e^{-t}$.

Complex Roots

Problem. $y'' + 4y = 0$, $y(0) = 2, y'(0) = 3$.
Transform: $(s^2 + 4)Y = 2s + 3$.
$$Y = \frac{2s}{s^2+4} + \frac{3}{s^2+4}$$
Answer: $y(t) = 2\cos(2t) + \frac{3}{2}\sin(2t)$.

Pattern

Each case reduces to factoring the characteristic polynomial in s, partial fractions, and table lookup. The Laplace method handles all three cases uniformly.

Forced Oscillation — Two Examples

Sinusoidal Forcing

Problem. $y'' + 4y = 3\cos t$, $y(0) = y'(0) = 0$.
Transform: $(s^2+4)Y = \frac{3s}{s^2+1}$.

$$Y = \frac{3s}{(s^2+1)(s^2+4)} = \frac{s}{s^2+1} - \frac{s}{s^2+4}$$

Answer: $y(t) = \cos t - \cos(2t)$.

Exponential Forcing

Problem. $y' - 2y = e^{5t}$, $y(0) = 3$.
Transform: $(s-2)Y - 3 = \frac{1}{s-5}$.

$$Y = \frac{3s-14}{(s-5)(s-2)} = \frac{1/3}{s-5} + \frac{8/3}{s-2}$$

Answer: $y(t) = \frac{1}{3}e^{5t} + \frac{8}{3}e^{2t}$.

Key Takeaway

Both examples reduce to factoring a polynomial in s^2 (or s), partial fractions, and table lookup. The Laplace method sidesteps the classical "particular + homogeneous" decomposition.

Discontinuous Forcing — Heaviside in Action

A Mass-Spring Switched Off at $t = 7$

Problem. $y'' + 2y' + 5y = h(t)$, $y(0) = y'(0) = 0$, where $h(t) = 5$ for $t < 7$ and 0 for $t > 7$.
Equivalently: $h(t) = 5(1 - u_7(t))$.
Step 1. Transform (using $\mathcal{L}\{1\} = 1/s$ and $\mathcal{L}\{u_7\} = e^{-7s}/s$):

$$(s^2 + 2s + 5)\, Y = \frac{5}{s} - \frac{5e^{-7s}}{s}$$

Step 2. Solve:

$$Y(s) = \frac{5}{s(s^2 + 2s + 5)} - \frac{5e^{-7s}}{s(s^2 + 2s + 5)}$$

A Mass-Spring Switched Off at $t = 7$ (cont.)

Step 3. Decompose the first term:

$$\frac{5}{s(s^2+2s+5)} = \frac{1}{s} - \frac{s+2}{(s+1)^2+4}$$

Complete the square: $(s+1)^2 + 4 = (s+1)^2 + 2^2$.

$$\mathcal{L}^{-1}\left\{\frac{s+2}{(s+1)^2+4}\right\} = e^{-t}\cos(2t) + \tfrac{1}{2}e^{-t}\sin(2t)$$

Step 4. Apply the second shifting property to the e^{-7s} term (delay by 7, zero before $t = 7$).
Answer:

$$\begin{aligned} y(t) &= 1 - e^{-t}\cos(2t) - \tfrac{1}{2}e^{-t}\sin(2t) \\ &\quad - u_7(t)\Big[1 - e^{-(t-7)}\cos(2(t-7)) - \tfrac{1}{2}e^{-(t-7)}\sin(2(t-7))\Big] \end{aligned}$$

Discontinuous Forcing — Heaviside in Action (cont.)

What Just Happened?

A single Laplace-transform equation handled *both regimes* of the forcing (on for $t < 7$, off for $t > 7$) in one algebraic expression. The classical method would require solving *two separate IVPs* and stitching the solutions by enforcing continuity at $t = 7$.

Common Pitfall

When applying the second shifting property to $e^{-7s}F(s)$, the inverse $f(t)$ must be expressed in terms of t, then replaced by $t \to t - 7$ and multiplied by $u_7(t)$. Forgetting the u_7 factor is the most common mistake.

Impulse Forcing — A Hammer Strike at $t = 4$

Damped Oscillator Struck at $t = 4$

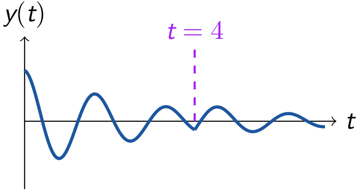
Problem. $y'' + 2y' + 26y = \delta_4(t)$, $y(0) = 1, y'(0) = 0$.

Step 1. Transform (using $\mathcal{L}\{\delta_4\} = e^{-4s}$):

$$[s^2 Y - s] + 2[sY - 2] + 26Y = e^{-4s}$$

Step 2. Solve:

$$Y(s) = \frac{s+2}{(s+1)^2 + 25} + \frac{e^{-4s}}{(s+1)^2 + 25}$$



Damped oscillation; new mode appears at $t = 4$

Key Takeaway

An impulse $\delta_a(t)$ produces $e^{-as}H(s)$ in $Y(s)$, which inverts to $u_a(t)\,h(t-a)$ — the impulse response delayed by a. The system "remembers" the impulse forever.

Impulse Forcing — A Hammer Strike at $t = 4$ (cont.)

Damped Oscillator Struck at $t = 4$ (cont.)

Step 3. Split the first term:

$$\frac{s+2}{(s+1)^2+25} = \frac{s+1}{(s+1)^2+5^2} + \frac{1}{5} \cdot \frac{5}{(s+1)^2+5^2}$$

Invert: $e^{-t}\cos(5t) + \frac{1}{5}e^{-t}\sin(5t)$.
Step 4. For the e^{-4s} term, apply the second shifting property:

$$\mathcal{L}^{-1}\left\{\frac{e^{-4s}}{(s+1)^2+25}\right\} = u_4(t) \cdot \tfrac{1}{5}e^{-(t-4)}\sin(5(t-4))$$

Answer:

$$\begin{aligned} y(t) = e^{-t}\cos(5t) &+ \tfrac{1}{5}e^{-t}\sin(5t) \\ &+ u_4(t) \cdot \tfrac{1}{5}e^{-(t-4)}\sin(5(t-4)) \end{aligned}$$

The third term is a new oscillation that switches on at $t = 4$ — the impulse response.

Periodic Functions and Their Transforms

Periodic Function Transform

If f has period P (i.e., $f(t+P) = f(t)$):

$$\mathcal{L}\{f(t)\} = \frac{1}{1-e^{-Ps}} \int_0^P e^{-st} f(t)\, dt$$

Proof Sketch

Split $\int_0^\infty = \int_0^P + \int_P^{2P} + \cdots$. By periodicity, the k-th integral equals $e^{-kPs}\int_0^P e^{-st} f(t)\, dt$. Sum the geometric series $\sum e^{-kPs} = 1/(1-e^{-Ps})$.

Square Wave

$f(t) = +1$ on $[0,2)$, -1 on $[2,4)$, period $P = 4$.

$$\int_0^4 e^{-st} f(t)\, dt = \int_0^2 e^{-st} dt - \int_2^4 e^{-st} dt = \frac{(1-e^{-2s})^2}{s}$$

$$F(s) = \frac{(1-e^{-2s})^2}{s(1-e^{-4s})} = \frac{1-e^{-2s}}{s(1+e^{-2s})}$$

using $1 - e^{-4s} = (1-e^{-2s})(1+e^{-2s})$.

Sawtooth Wave

$f(t) = t$ on $[0,1)$, period $P = 1$. After integration by parts:

$$F(s) = \frac{1}{s^2} - \frac{e^{-s}}{s(1-e^{-s})}$$

Solving Linear Systems with Laplace

The Method

For a system

$$a_1 x' + a_2 y' + a_3 x + a_4 y = \beta_1(t)$$
$$b_1 x' + b_2 y' + b_3 x + b_4 y = \beta_2(t)$$

1. Transform each equation: $x' \to sX - x(0)$, $y' \to sY - y(0)$, $x \to X$, $y \to Y$.
2. The result is a 2×2 linear algebraic system in $X(s), Y(s)$.
3. Solve by elimination or Cramer's rule.
4. Apply partial fractions and invert

Worked Example

System: $x' - 6x + 3y = 8e^t$, $y' - 2x - y = 4e^t$, with $x(0) = -1, y(0) = 0$.
Transform both equations:

$$(sX + 1) - 6X + 3Y = \frac{8}{s-1}$$
$$sY - 2X - Y = \frac{4}{s-1}$$

Rearrange:

$$(s-6)X + 3Y = \frac{-s+9}{s-1}$$
$$-2X + (s-1)Y = \frac{4}{s-1}$$

Solving Linear Systems with Laplace (cont.)

Worked Example (cont.)

Solve (Cramer's rule):

$$X(s) = \frac{-s+7}{(s-1)(s-4)}, \quad Y(s) = \frac{2}{(s-1)(s-4)}$$

Partial fractions and invert:

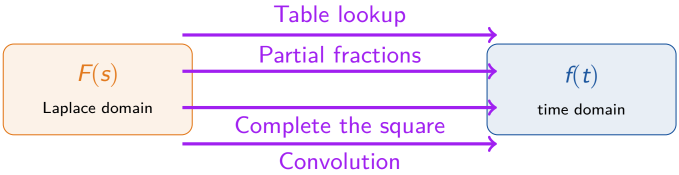
$$x(t) = -2e^{t} + e^{4t}, \quad y(t) = -\tfrac{2}{3}e^{t} + \tfrac{2}{3}e^{4t}$$

Solving Linear Systems with Laplace (cont.)

Key Takeaway

The Laplace method scales effortlessly to systems — the algebraic structure in s is what would be obtained by diagonalizing the coefficient matrix, but the ICs are incorporated from the start.

Inverse Laplace — The Four Techniques



Technique 1 — Table Lookup

Memorize the six starter transforms. Add shifting properties for $(s-a)$ patterns, derivative rules for ODEs. Most problems reduce to this.

Example: $\mathcal{L}^{-1}\left\{\frac{3}{s^2+9}\right\} = \sin(3t)$.

Technique 2 — Partial Fractions

For rational $F(s) = N(s)/D(s)$ with $\deg N < \deg D$, factor $D(s)$ and decompose. Distinct linear $\to \sum A_k/(s-r_k)$; repeated $\to \sum A_k/(s-r)^k$; irreducible quadratics $\to (Bs+C)/(s^2+bs+c)$.

Example: $\frac{3s-14}{(s-5)(s-2)} = \frac{1/3}{s-5} + \frac{8/3}{s-2} \to \frac{1}{3}e^{5t} + \frac{8}{3}e^{2t}$.

Inverse Laplace — The Four Techniques (cont.)

Technique 3 — Complete the Square

For irreducible $s^2 + bs + c$, rewrite as $(s + b/2)^2 + (c - b^2/4)$. Match to $\frac{s+a}{(s+a)^2+\omega^2}$ (cosine) or $\frac{\omega}{(s+a)^2+\omega^2}$ (sine); first shifting gives e^{-at}.

Example: $\frac{1}{s^2+6s+13} = \frac{1}{(s+3)^2+4} \to \frac{1}{2}e^{-3t}\sin(2t)$.

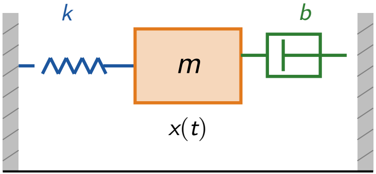
Technique 4 — Convolution

For products $F(s)G(s)$ resisting partial fractions:

$$\mathcal{L}^{-1}\{FG\} = f * g = \int_0^t f(\tau)\, g(t-\tau)\, d\tau$$

Example: $\mathcal{L}^{-1}\left\{\frac{1}{s^2(s+1)}\right\} = t * e^{-t} = t + e^{-t} - 1$.

The Mass-Spring-Dashpot System



The ODE

$$mx'' + bx' + kx = f(t)$$

$$x(0) = x_0, \quad x'(0) = v_0$$

General Solution via Laplace

Transform: $m(s^2X - sx_0 - v_0) + b(sX - x_0) + kX = F(s)$. Solve:

$$X(s) = \frac{F(s) + m(x_0 s + v_0) + b\,x_0}{ms^2 + bs + k}$$

The denominator $ms^2 + bs + k$ is the *characteristic polynomial*. Its roots determine the homogeneous behavior (overdamped, critically damped, underdamped).

The Mass-Spring-Dashpot System (cont.)

Forced Undamped Motion

$x'' + 4x = \sin(3t)$, $x(0) = x'(0) = 0$. Transform:

$$(s^2+4)X = \frac{3}{s^2+9} \Rightarrow X = \frac{3}{(s^2+4)(s^2+9)}$$

Partial fractions:

$$\frac{3}{(s^2+4)(s^2+9)} = \frac{3/5}{s^2+4} - \frac{3/5}{s^2+9}$$

Invert: $x(t) = \frac{3}{10}\sin(2t) - \frac{1}{5}\sin(3t)$.
A superposition of two oscillations — natural frequency 2 and forcing frequency 3. No resonance.

Key Takeaway

The Laplace method gives the full solution — homogeneous + particular + ICs — in one algebraic equation. The denominator reveals the natural frequencies; the numerator reveals the forcing response.

PART IV
Convolution & Systems

The convolution theorem. Duhamel's principle. Volterra equations. Transfer functions. Impulse response. Poles, zeros, and stability.

The Convolution Theorem

Definition and Theorem

Convolution:

$(f * g)(t) = \int_0^t f(\tau)\, g(t-\tau)\, d\tau = \int_0^t f(t-\tau)\, g(\tau)\, d\tau$

The two forms are equal (substitute $u = t - \tau$).

Convolution Theorem:

$$\mathcal{L}\{f * g\} = \mathcal{L}\{f\} \cdot \mathcal{L}\{g\} = F(s)\, G(s)$$

$$\mathcal{L}^{-1}\{F(s)\, G(s)\} = (f * g)(t)$$

Proof Sketch — the geometric version

Write the product $F(s)\, G(s)$ as a double integral over the first quadrant:

$$F(s)\, G(s) = \int_0^\infty \int_0^\infty e^{-s(u+v)} f(u)\, g(v)\, du\, dv$$

Change variables: $t = u + v$, $\tau = v$ (Jacobian $= 1$). The first quadrant becomes the wedge $0 \leq \tau \leq t < \infty$. Swap the order of integration: $F(s)\, G(s) = \int_0^\infty e^{-st} \left[\int_0^t f(t-\tau)\, g(\tau)\, d\tau\right] dt = \mathcal{L}\{f * g\}$

Key Takeaway

The convolution theorem is the *product rule* for Laplace transforms. It converts products in the s-domain into convolutions in the t-domain. This is the foundation for transfer functions and Duhamel's principle.

Convolution: Properties and Pitfalls

Algebraic Properties

For piecewise-continuous exponentially-bounded f, g, h and constant c:

- **Commutative:** $f * g = g * f$
- **Associative:** $f * (g * h) = (f * g) * h$
- **Distributive:** $f * (g + h) = f * g + f * h$
- **Scalar:** $(cf) * g = c(f * g)$
- **Zero:** $0 * f = f * 0 = 0$

CRITICAL: No Multiplicative Identity

$f * 1 \neq f$. The function 1 is *not* the identity for convolution. For example, $(\cos t) * 1 = \sin t$, *not* $\cos t$.

The identity for convolution is the Dirac delta: $(f * \delta)(t) = f(t)$. This is the sifting property.

Pitfall 2 — $f * f$ not non-negative

Unlike $f(t)^2$, the convolution $f * f$ can be negative. For example, $(\sin t) * (\sin t) = \frac{1}{2}(\sin t - t\cos t)$, which is negative on intervals.

Convolution: Properties and Pitfalls (cont.)

$(\cos t) * (\cos t)$

Use $\cos(t-\tau)\cos\tau = \frac{1}{2}[\cos(t-2\tau)+\cos t]$:

$$
\begin{aligned}
(\cos * \cos)(t) &= \int_0^t \cos(t-\tau)\cos\tau \, d\tau \\
&= \frac{1}{2}\int_0^t [\cos(t-2\tau)+\cos t] \, d\tau \\
&= \frac{1}{2}[\sin t + t\cos t]
\end{aligned}
$$

Duhamel's Principle — Solution for Arbitrary Input

The Killer Application

Problem. Solve $x'' + \omega_0^2 x = f(t)$, $x(0) = x'(0) = 0$, for *arbitrary* $f(t)$.

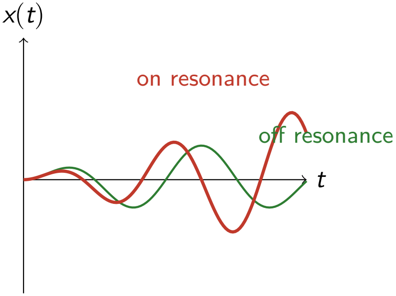
Step 1. Transform: $s^2 X + \omega_0^2 X = F(s)$, so

$$X(s) = \frac{F(s)}{s^2 + \omega_0^2}$$

Step 2. Recognize $\frac{1}{s^2+\omega_0^2}$ as the *transfer function* $H(s)$, with inverse $h(t) = \frac{\sin(\omega_0 t)}{\omega_0}$.

Step 3. Apply the convolution theorem: $X(s) = H(s)\,F(s)$ implies

$$x(t) = (h * f)(t) = \int_0^t \frac{\sin(\omega_0(t-\tau))}{\omega_0}\, f(\tau)\, d\tau$$



Resonance: amplitude grows linearly

Key Takeaway

Once the impulse response $h(t)$ of a system is known, its response to *any* input is the convolution with h. The system is fully characterized by h.

Duhamel's Principle — Solution for Arbitrary Input (cont.)

The Killer Application (cont.)

Step 4. Resonance check: if $f(t) = \cos(\omega_0 t)$, the convolution evaluates to

$$x(t) = \frac{t\,\sin(\omega_0 t)}{2\omega_0}$$

— the linearly-growing amplitude that defines *resonance*.

Volterra Integral Equations

The Volterra Equation

$$x(t) = f(t) + \int_0^t g(t-\tau)\,x(\tau)\,d\tau$$

The unknown x appears under the integral — but the integral is a convolution $(g * x)(t)$. Apply $\mathcal{L}$:

$$X(s) = F(s) + G(s)\,X(s)$$

$$\Rightarrow\ X(s) = \frac{F(s)}{1 - G(s)}$$

Pure algebra. Then invert $X(s)$.

Worked Example

Equation: $x(t) = e^{-t} + \int_0^t \sinh(t-\tau)\,x(\tau)\,d\tau$.
Transform (using $\mathcal{L}\{e^{-t}\} = \frac{1}{s+1}$ and $\mathcal{L}\{\sinh t\} = \frac{1}{s^2-1}$):

$$X = \frac{1}{s+1} + \frac{1}{s^2-1}X$$

Solve:

$$X\left(1 - \frac{1}{s^2-1}\right) = \frac{1}{s+1} \ \Rightarrow\ X \cdot \frac{s^2-2}{s^2-1} = \frac{1}{s+1}$$

$$X = \frac{s^2-1}{(s+1)(s^2-2)} = \frac{s-1}{s^2-2}$$

using $s^2 - 1 = (s-1)(s+1)$.

Volterra Integral Equations (cont.)

Worked Example (cont.)

Decompose: $\frac{s-1}{s^2-2} = \frac{s}{s^2-2} - \frac{1}{s^2-2}$. Recognize:

$$\mathcal{L}^{-1}\left\{\frac{s}{s^2-2}\right\} = \cosh(\sqrt{2}\,t), \quad \mathcal{L}^{-1}\left\{\frac{1}{s^2-2}\right\} = \frac{1}{\sqrt{2}}\sinh(\sqrt{2}\,t)$$

Answer: $x(t) = \cosh(\sqrt{2}\,t) - \frac{1}{\sqrt{2}}\sinh(\sqrt{2}\,t)$.

Why This Matters

Integral equations arise in physics problems with memory: viscoelastic materials, population models with age structure, radiative transfer. Laplace transforms solve them as cleanly as differential equations.

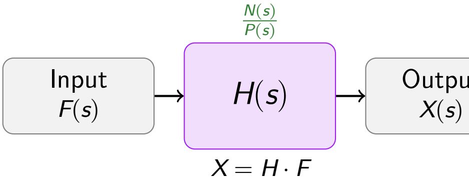
Transfer Functions $H(s)$

Definition

For a linear constant-coefficient ODE $P(D)\,x = N(D)\,E(t)$ with *rest* initial conditions ($x(0) = x'(0) = \cdots = 0$):

$$H(s) = \frac{\mathcal{L}\{x\}}{\mathcal{L}\{E\}} = \frac{N(s)}{P(s)} = \frac{b_m s^m + \cdots + b_0}{a_n s^n + \cdots + a_0}$$

$H(s)$ depends *only on the system*, not on the input. Once H is known, the output for any input is $X(s) = H(s)\,E(s)$.



Key Takeaway

A system *is* its transfer function. Two systems with the same $H(s)$ are indistinguishable from their input-output behavior — even if their internal structures differ.

Transfer Functions $H(s)$ (cont.)

Why It Matters

- Characterizes the system independent of input.
- Poles of H (roots of P) determine natural frequencies and stability.
- Zeros of H (roots of N) determine which inputs are "absorbed."
- Engineers measure $H(i\omega)$ (frequency response) experimentally.

Mass-Spring with Base Excitation

For $mx'' + bx' + kx = bE'(t) + kE(t)$:

$$H(s) = \frac{bs + k}{ms^2 + bs + k}$$

Poles at $s = \frac{-b \pm \sqrt{b^2 - 4mk}}{2m}$; zero at $s = -k/b$.

The Impulse Response $h(t)$

Definition

$$h(t) = \mathcal{L}^{-1}\{H(s)\}$$

— the inverse Laplace transform of the transfer function. Equivalently: the solution to $Lh = \delta(t)$ with rest initial conditions.

The equivalence holds because $\mathcal{L}\{\delta\} = 1$, so $X(s) = H(s) \cdot 1 = H(s)$, hence $x(t) = h(t)$.

Undamped Oscillator

Problem. Find the impulse response of $x'' + \omega_0^2 x = \delta(t)$, $x(0) = x'(0) = 0$.
Transform: $(s^2 + \omega_0^2)X = 1$, so $X(s) = \frac{1}{s^2+\omega_0^2} = H(s)$. Invert:

$$h(t) = \frac{\sin(\omega_0 t)}{\omega_0}$$

The Impulse Response $h(t)$ (cont.)

The Big Payoff

For any input $f(t)$, the output is the convolution with h:

$$x(t) = (h * f)(t) = \int_0^t h(t-\tau)\, f(\tau)\, d\tau$$

This is the Duhamel / superposition formula. The system is fully characterized by its impulse response.

Undamped Oscillator (cont.)

Now use this h to solve $x'' + \omega_0^2 x = f(t)$ for *any* f:

$$x(t) = (h * f)(t) = \int_0^t \frac{\sin(\omega_0(t-\tau))}{\omega_0}\, f(\tau)\, d\tau$$

A single integral — no ODE solving required.

The Impulse Response $h(t)$ (cont.)

Indicial vs Impulse Response

The *indicial response* $A(t)$ (response to unit step $u(t)$) and the *impulse response* $h(t)$ (response to $\delta(t)$) are related by $A'(t) = h(t)$, i.e., $A(t) = \int_0^t h(\tau)\, d\tau$. Both characterize the system; h is more fundamental.

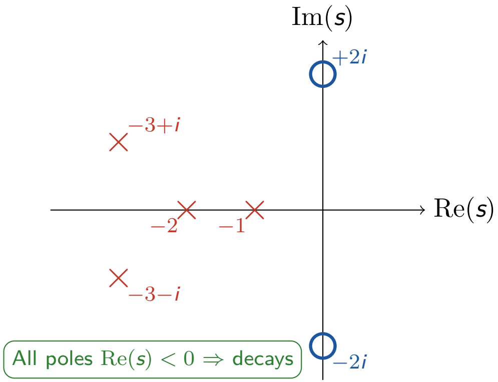
Poles, Zeros, and Stability

Definitions

For $H(s) = N(s)/P(s)$ in *reduced form*:

- **Poles:** roots of $P(s)$ — where $H \to \infty$.
- **Zeros:** roots of $N(s)$ — where $H = 0$.

The poles are exactly the roots of the characteristic equation — they determine the natural frequencies.



Pole-zero plot: $H(s) = \frac{s^2+4}{(s^2+3s+2)(s^2+6s+10)}$

Poles, Zeros, and Stability (cont.)

Stability Theorem

For the homogeneous solution $x_h(t)$:

- $\mathrm{Re}(s_i) < 0$ for all poles $\Rightarrow x_h \to 0$ exponentially (stable).
- $\mathrm{Re}(s_i) > 0$ for any pole $\Rightarrow x_h$ grows (unstable).
- $\mathrm{Im}(s_i) \neq 0$ for any pole $\Rightarrow$ oscillations.
- The rightmost pole dominates long-term behavior.

Common Pitfall

A zero at $s = s_z$ means an input $e^{s_z t}$ produces no long-term response — the system "absorbs" it. This is the principle behind notch filters.

Key Takeaway

The pole-zero plot is a complete picture of the system's natural behavior. Left half-plane $\to$ stable; imaginary axis $\to$ marginal; right half-plane $\to$ unstable.

PART V
Advanced Applications

PDEs via Laplace. Beam bending. Abel's tautochrone.
Real-integral tricks. Resonance. The transform's reach beyond ODEs.

The Transport Equation via Laplace

River Pollution — Wavefront at Speed α

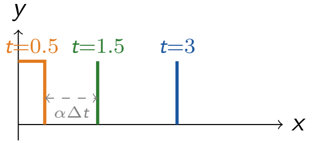
PDE. $y_t = -\alpha\, y_x$ for $x > 0, t > 0$, with $y(0,t) = C$ and $y(x,0) = 0$. Here $\alpha > 0$ is the wave speed.

Physical setup: a river carrying toxic waste from a factory at $x = 0$. The concentration at the factory is held at C; the river starts clean.

Step 1. Transform in t (treat x as a parameter). The PDE becomes an ODE in x:

$$s\,Y(x) = -\alpha\, Y'(x)$$

Step 2. Transform the BC $y(0,t) = C$: $Y(0) = C/s$.



Wavefront travels at speed α

Key Takeaway

The Laplace transform reduces a PDE in two variables to an ODE in one by transforming one variable. The BC becomes the ODE's initial condition.

The Transport Equation via Laplace (cont.)

River Pollution (cont.)

Step 3. Solve the ODE:

$$Y(x) = \frac{C}{s}\, e^{-sx/\alpha}$$

Step 4. Invert using the second shifting property ($\mathcal{L}^{-1}\{\frac{1}{s}e^{-as}\} = u_a(t)$ with $a = x/\alpha$):

$$y(x,t) = C \cdot u(t - x/\alpha)$$

Half-Line with Prescribed Flux

PDE. $y_t = y_{xx}$ for $x > 0, t > 0$, with $y_x(0,t) = f(t)$ (prescribed heat flux) and $y(x,0) = 0$, with y bounded as $x \to \infty$.

Step 1. Transform in t: $sY = Y''$ — an ODE in x.

Step 2. General solution: $Y(x) = Ae^{\sqrt{s}\,x} + Be^{-\sqrt{s}\,x}$. Boundedness as $x \to \infty$ forces $A = 0$. So $Y(x) = B\,e^{-\sqrt{s}\,x}$.

Step 3. Apply the flux BC: $Y'(0) = -\sqrt{s}\,B = F(s)$, so $B = -F(s)/\sqrt{s}$. Hence

$$Y(x) = -\frac{F(s)}{\sqrt{s}}\,e^{-\sqrt{s}\,x}$$

The Heat Equation on a Half-Line (cont.)

Half-Line with Prescribed Flux (cont.)

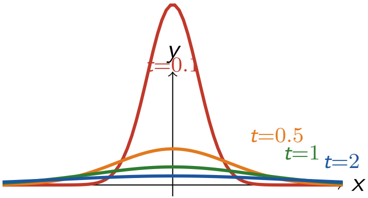
Step 4. Use the table identity:

$$\mathcal{L}^{-1}\left\{\frac{1}{\sqrt{s}}\, e^{-\sqrt{s}\,x}\right\} = \frac{1}{\sqrt{\pi t}}\, e^{-x^2/(4t)}$$

This is the *heat kernel* — the Green's function for diffusion.
Step 5. By linearity, the solution is the convolution of $-f$ with the heat kernel:

$$y(x,t) = -\int_0^t \frac{1}{\sqrt{\pi(t-\tau)}}\, e^{-x^2/(4(t-\tau))}\, f(\tau)\, d\tau$$

The Heat Equation on a Half-Line (cont.)



Heat kernel: spreading Gaussian as t increases

Key Takeaway

The Laplace transform handles nonhomogeneous boundary conditions natively — separation of variables struggles with them. The price: the answer is a convolution integral, not a closed form.

Three-Point Beam Bending with δ

Euler-Bernoulli Beam with Point Load

Equation. Euler-Bernoulli beam with point load F at $x = a$:

$$EI\,\frac{d^4y}{dx^4} = -F\,\delta(x-a), \quad 0 < x < L$$

Simply-supported BCs: $y(0) = y(L) = 0$ and $y''(0) = y''(L) = 0$.

Specific case: $L = 2$, $EI = 1$, $F = 1$, $a = 1$. So $y'''' = -\delta(x-1)$.
Step 1. Transform in x (treat x as the time-like variable):

$$s^4 Y - s^3 y(0) - s^2 y'(0) - s\,y''(0) - y'''(0) = -e^{-s}$$

Apply $y(0) = 0$ and $y''(0) = 0$. Let $C_1 = y'(0)$, $C_2 = y'''(0)$:

$$Y(s) = \frac{-e^{-s}}{s^4} + \frac{C_1}{s^2} + \frac{C_2}{s^4}$$

Three-Point Beam Bending with δ (cont.)

Euler-Bernoulli Beam — Inversion and BCs

Step 2. Invert using the second shifting property (for e^{-s}) and table lookup:

$$y(x) = -\frac{(x-1)^3}{6}\,u(x-1) + C_1\,x + \frac{C_2\,x^3}{6}$$

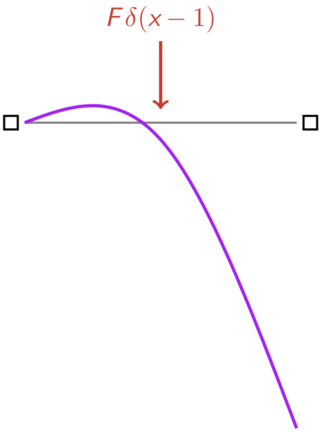
Step 3. Apply the right-end BCs $y(2) = 0$ and $y''(2) = 0$ (here $u(x-1) = 1$):

$$y(2) = -\tfrac{1}{6} + 2C_1 + \tfrac{4C_2}{3} = 0$$
$$y''(2) = -1 + 2C_2 = 0 \;\Rightarrow\; C_2 = \tfrac{1}{2}$$

Solving: $C_1 = -\frac{1}{4}$.

Answer: $y(x) = -\frac{(x-1)^3}{6}\,u(x-1) - \frac{x}{4} + \frac{x^3}{12}$.

Three-Point Beam Bending with δ (cont.)



Key Takeaway

Point loads on beams are naturally modeled by $\delta(x-a)$. The Laplace transform (in the spatial variable x) handles them exactly — no need to split the beam into segments and match BCs at the load point.

Abel's Problem and the Cycloid

The Historical Jewel

Setup. A bead slides frictionlessly down a wire from height y to the origin under gravity. Abel's problem: given the descent-time function $T(y)$, find the wire's shape.
Conservation of energy gives $dt = -ds/\sqrt{2g(y-v)}$. Integrating from $v = y$ to $v = 0$:

$$T(y) = \frac{1}{\sqrt{2g}} \int_0^y \frac{f(v)}{\sqrt{y-v}}\, dv$$

where $f(v) = \sqrt{1 + (dx/dy)^2}$. This is a *convolution*: $T = \frac{1}{\sqrt{2g}}(y^{-1/2} * f)$.
Apply $\mathcal{L}$ and use $\mathcal{L}\{y^{-1/2}\} = \sqrt{\pi/s}$:

$$\mathcal{L}\{T\} = \frac{1}{\sqrt{2g}}\sqrt{\frac{\pi}{s}}\,\mathcal{L}\{f\} \;\Rightarrow\; \mathcal{L}\{f\} = \sqrt{\frac{2g}{\pi}}\, s^{-1/2}\,\mathcal{L}\{T\}$$

Abel's Problem and the Cycloid (cont.)

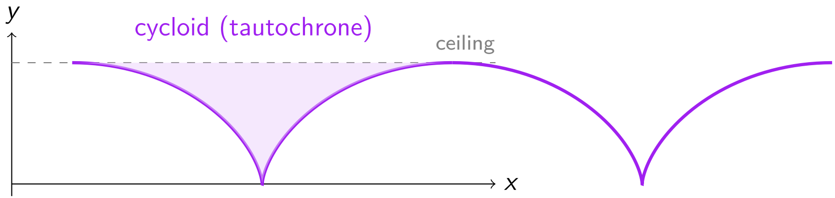
The Historical Jewel (cont.)

Tautochrone. Set $T(y) = T_0$ (constant — same descent time from any height). Then $\mathcal{L}\{f\} = \sqrt{2g/\pi}\, T_0\, s^{-3/2}$, which inverts to $f(y) = b/\sqrt{y}$ with $b = \sqrt{2g}\, T_0/\pi$.
Solve $(dx/dy)^2 = (b - y)/y$. Parametrize $y = b\sin^2\phi$ and integrate. With $a = b/2$ and $\theta = 2\phi$:

$$x = a(\theta + \sin\theta), \quad y = a(1 - \cos\theta)$$

This is a cycloid — the curve traced by a point on a rolling circle. Huygens (1673) discovered the tautochrone property geometrically; Abel (1826) derived it analytically via this integral-equation approach *without knowing the answer in advance.*

Abel's Problem and the Cycloid (cont.)



Key Takeaway

Abel's problem reduces to a convolution integral equation $T = \frac{1}{\sqrt{2g}}(y^{-1/2} * f)$, which the Laplace transform solves algebraically. The tautochrone is a cycloid — a result Huygens found geometrically, Abel found analytically.

Evaluating Real Integrals via Laplace

The Trick

From the integration-in-s rule $\mathcal{L}\{f(t)/t\} = \int_s^\infty F(u)\,du$, take $s \to 0^+$:

$$\int_0^\infty \frac{f(t)}{t}\,dt = \int_0^\infty F(u)\,du$$

If the RHS is computable from a known transform $F = \mathcal{L}\{f\}$, a hard real integral is evaluated.

Dirichlet Integral

Use $\mathcal{L}\{\sin t\} = \frac{1}{s^2+1}$. Then $\int_0^\infty \frac{\sin t}{t}\,dt = \int_0^\infty \frac{1}{u^2+1}\,du = \arctan(u)\Big|_0^\infty = \frac{\pi}{2}$

The famous Dirichlet integral in two lines.

Frullani-Type Integral

Use $\mathcal{L}\{e^{-at}\} = \frac{1}{s+a}$. Then $\int_0^\infty \frac{e^{-at}-e^{-bt}}{t}\,dt = \int_0^\infty \left[\frac{1}{u+a} - \frac{1}{u+b}\right] du = \ln \frac{b}{a}$

Damped Sine Integral

Use $\mathcal{L}\{e^{-at}\sin(bt)\} = \frac{b}{(s+a)^2+b^2}$. Then

$$\int_0^\infty \frac{e^{-at}\sin(bt)}{t}\,dt = \arctan\frac{b}{a}$$

Evaluating Real Integrals via Laplace (cont.)

Key Takeaway

The Laplace transform is not just for ODEs — it is a general integration tool. Any time a $1/t$ factor appears in a real integral, consider $\mathcal{L}\{f(t)/t\} = \int_s^\infty F$.

Resonance Emerges Naturally from Convolution

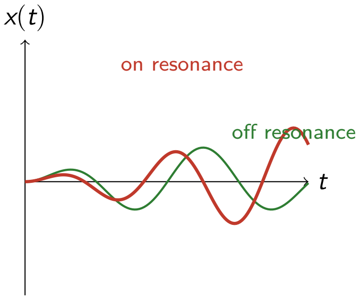
Driving at the Natural Frequency

Consider $x'' + \omega_0^2 x = \cos(\omega_0 t)$, $x(0) = x'(0) = 0$ — drive at the natural frequency.
From Duhamel's principle:

$$x(t) = \int_0^t \frac{\sin(\omega_0(t-\tau))}{\omega_0} \cos(\omega_0 \tau)\, d\tau$$

Use the product-to-sum identity $\sin A \cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$:
$\sin(\omega_0(t-\tau))\cos(\omega_0\tau) = \frac{1}{2}[\sin(\omega_0 t) + \sin(\omega_0(t-2\tau))]$
Integrate:

$$x(t) = \frac{1}{2\omega_0}\int_0^t [\sin(\omega_0 t) + \sin(\omega_0(t-2\tau))]\, d\tau$$
$$= \frac{1}{2\omega_0}\left[t\sin(\omega_0 t) + \frac{\cos(\omega_0(t-2\tau))}{2\omega_0}\bigg|_0^t\right]$$
$$= \frac{t\sin(\omega_0 t)}{2\omega_0}$$



Key Takeaway

Resonance is not a special case to be memorized. It falls out of the convolution integral naturally when the forcing frequency matches a natural frequency — the t factor in front of the sine is the signature.

PART VI
Reference Tables & Examples

Master Laplace and inverse-Laplace tables. Encyclopedic worked examples. Historical sketches. Cheat-sheet summary. Closing.

Master Table of Laplace Transforms

$f(t)$	$F(s)$	**Valid**	$f(t)$	$F(s)$	**Valid**
1	$\frac{1}{s}$	$s>0$	$u_a(t)$	$\frac{e^{-as}}{s}$	$s>0$
t	$\frac{1}{s^2}$	$s>0$	$u_a(t)f(t-a)$	$e^{-as}F(s)$	$s>\alpha$
t^n	$\frac{n!}{s^{n+1}}$	$s>0$	$e^{at}f(t)$	$F(s-a)$	$s>\alpha+a$
t^α	$\frac{\Gamma(\alpha+1)}{s^{\alpha+1}}$	$s>0$	$f(ct)$	$\frac{1}{c}F(s/c)$	$c>0$
e^{at}	$\frac{1}{s-a}$	$s>a$	$\delta(t-t_0)$	e^{-st_0}	all s
$\sin(at)$	$\frac{a}{s^2+a^2}$	$s>0$	$f'(t)$	$sF-f(0)$	$s>\alpha$
$\cos(at)$	$\frac{s}{s^2+a^2}$	$s>0$	$f''(t)$	$s^2F-sf(0)-f'(0)$	$s>\alpha$
$\sinh(at)$	$\frac{a}{s^2-a^2}$	$s>\|a\|$	$f^{(n)}(t)$	$s^nF-\sum s^{n-1-k}f^{(k)}(0)$	$s>\alpha$
$\cosh(at)$	$\frac{s}{s^2-a^2}$	$s>\|a\|$	$\int_0^t f\,d\tau$	$\frac{F(s)}{s}$	$s>\alpha$
$e^{at}\sin(bt)$	$\frac{b}{(s-a)^2+b^2}$	$s>a$	$(f*g)(t)$	$F(s)\,G(s)$	$s>\max(\alpha,\beta)$
$e^{at}\cos(bt)$	$\frac{s-a}{(s-a)^2+b^2}$	$s>a$	$t\sin(at)$	$\frac{2as}{(s^2+a^2)^2}$	$s>0$
$t^n e^{at}$	$\frac{n!}{(s-a)^{n+1}}$	$s>a$	$t\cos(at)$	$\frac{s^2-a^2}{(s^2+a^2)^2}$	$s>0$

Usage

Recognize the s-domain pattern, then apply the corresponding rule. Shifting for $(s-a)$ patterns; second shifting for e^{-as} factors; convolution for products $F(s)G(s)$.

Master Table of Inverse Laplace Transforms

$F(s)$ pattern	$f(t)$	Technique	$F(s)$ pattern	$f(t)$	Technique
$\frac{1}{s}$	1	Direct	$\frac{e^{-as}}{s}$	$u_a(t)$	2nd shift
$\frac{1}{s^n}$	$\frac{t^{n-1}}{(n-1)!}$	Direct	$\frac{e^{-as}}{s-a}$	$u_a(t)\, e^{a(t-a)}$	2nd shift
$\frac{1}{s-a}$	e^{at}	Direct	$\frac{b\, e^{-as}}{(s-a)^2+b^2}$	$u_a\, e^{a(t-a)} \sin b(t-a)$	2nd shift
$\frac{a}{s^2+a^2}$	$\sin(at)$	Direct	$e^{-as} F(s)$	$u_a(t)\, f(t-a)$	2nd shift (gen.)
$\frac{s}{s^2+a^2}$	$\cos(at)$	Direct	$\frac{F(s)}{s}$	$\int_0^t f\, d\tau$	Integration
$\frac{a}{s^2-a^2}$	$\sinh(at)$	Direct	$F(s)\, G(s)$	$(f * g)(t)$	Convolution
$\frac{s}{s^2-a^2}$	$\cosh(at)$	Direct	1	$\delta(t)$	Delta
$\frac{n!}{(s-a)^{n+1}}$	$t^n e^{at}$	Direct	e^{-as}	$\delta(t-a)$	Delta
$\frac{b}{(s-a)^2+b^2}$	$e^{at} \sin(bt)$	1st shift	$-F'(s)$	$t\, f(t)$	Diff. in s
$\frac{s-a}{(s-a)^2+b^2}$	$e^{at} \cos(bt)$	1st shift	$\int_s^\infty F\, du$	$\frac{f(t)}{t}$	Integ. in s
$\frac{1}{(s-a)(s-b)}$	$\frac{e^{at}-e^{bt}}{a-b}$	Partial frac.	$\frac{1}{\sqrt{s^2+a^2}}$	$J_0(at)$	Bessel
$\frac{1}{(s-a)^2}$	$t\, e^{at}$	Partial frac.	$\frac{As+B}{(s-a)^2+b^2}$	$e^{at}[C_1 \sin bt + C_2 \cos bt]$	Comp. square

Usage

Inverse-Laplace lookup works by pattern matching. See $(s-a)^2 + b^2 \to$ first shifting; see $e^{-as} \to$ second shifting; see $F(s)G(s) \to$ convolution.

Worked Examples — Forward Transforms

Polynomial × Exponential

Problem. Find $\mathcal{L}\{t^2 e^{-3t}\}$.
Use $\mathcal{L}\{t^n e^{at}\} = \frac{n!}{(s-a)^{n+1}}$ with $n = 2, a = -3$:
$\mathcal{L}\{t^2 e^{-3t}\} = \frac{2!}{(s+3)^3} = \frac{2}{(s+3)^3}, \quad s > -3$

Trig via Euler

Problem. Find $\mathcal{L}\{\sin(3t)\cos(5t)\}$.
Use $\sin A \cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$:

$$\sin(3t)\cos(5t) = \tfrac{1}{2}[\sin(8t) - \sin(2t)]$$

$$\mathcal{L}\{\cdots\} = \frac{1}{2}\left[\frac{8}{s^2+64} - \frac{2}{s^2+4}\right]$$

Hyperbolic via Linearity

Problem. Find $\mathcal{L}\{\cosh(at)\}$.
Use $\cosh(at) = \frac{e^{at}+e^{-at}}{2}$:
$\mathcal{L}\{\cosh(at)\} = \frac{1}{2}\left[\frac{1}{s-a} + \frac{1}{s+a}\right] = \frac{s}{s^2-a^2}$
for $s > |a|$.

Piecewise Function

Problem. Find $\mathcal{L}\{f\}$ where $f(t) = 0$ for $t < 1$, $t-1$ for $1 < t < 3$, 2 for $t > 3$.
Write $f(t) = (t-1)u_1(t) - (t-3)u_3(t)$. By the second shifting property:

$$\mathcal{L}\{f\} = \frac{e^{-s}}{s^2} - \frac{e^{-3s}}{s^2} = \frac{e^{-s} - e^{-3s}}{s^2}$$

Worked Examples — Inverse Transforms

Distinct Linear Factors

Problem. $\mathcal{L}^{-1}\left\{\frac{3s-14}{(s-5)(s-2)}\right\}$

Partial fractions: $\frac{A}{s-5}+\frac{B}{s-2}$. Solve: $A=\frac{1}{3}, B=\frac{8}{3}$.

$$\mathcal{L}^{-1}\{\cdots\}=\frac{1}{3}e^{5t}+\frac{8}{3}e^{2t}$$

Delayed (Heaviside)

Problem. $\mathcal{L}^{-1}\left\{\frac{e^{-4s}}{s^2+9}\right\}$

Without e^{-4s}: $\mathcal{L}^{-1}\left\{\frac{1}{s^2+9}\right\}=\frac{1}{3}\sin(3t)$.

With e^{-4s}, apply second shifting:

$$\mathcal{L}^{-1}\{\cdots\}=u_4(t)\cdot\frac{1}{3}\sin(3(t-4))$$

A sine wave switching on at $t=4$.

Worked Examples — Inverse Transforms (cont.)

Irreducible Quadratic

Problem. $\mathcal{L}^{-1}\left\{\frac{2s+3}{s^2+2s+5}\right\}$

Complete the square: $s^2+2s+5=(s+1)^2+4$. Rewrite numerator: $2s+3=2(s+1)+1$.

$$\frac{2(s+1)}{(s+1)^2+2^2}+\frac{1}{2}\cdot\frac{2}{(s+1)^2+2^2}$$

Apply first shifting:

$$\mathcal{L}^{-1}\{\cdots\}=2e^{-t}\cos(2t)+\tfrac{1}{2}e^{-t}\sin(2t)$$

Convolution

Problem. $\mathcal{L}^{-1}\left\{\frac{1}{s^2(s+1)}\right\}$

Write as product: $\frac{1}{s^2}\cdot\frac{1}{s+1}$. Inverses: t and e^{-t}. Convolve: $\mathcal{L}^{-1}\{\cdots\}=(t)*(e^{-t})=\int_0^t(t-\tau)\,e^{-\tau}\,d\tau=t+e^{-t}-1$

Worked Examples — Solving IVPs

Simple Homogeneous

Problem. $y'' - 4y' + 4y = 0$, $y(0) = 1, y'(0) = 0$.
Transform: $(s-2)^2 Y = s$. Rewrite: $s = (s-2) + 2$.

$$Y = \frac{1}{s-2} + \frac{2}{(s-2)^2}$$

Answer: $y(t) = (1+2t)\, e^{2t}$.

Step Forcing

Problem. $y' + y = u_3(t)$, $y(0) = 1$.
Transform: $(s+1)Y = 1 + \frac{e^{-3s}}{s}$.
$Y = \frac{1}{s+1} + \frac{e^{-3s}}{s(s+1)} = \frac{1}{s+1} + e^{-3s}\left(\frac{1}{s} - \frac{1}{s+1}\right)$
Answer: $y(t) = e^{-t} + u_3(t)[1 - e^{-(t-3)}]$.

Worked Examples — Solving IVPs (cont.)

Forced Oscillation

Problem. $y'' + 9y = \sin(2t)$, $y(0) = y'(0) = 0$.
Transform: $(s^2 + 9)Y = \frac{2}{s^2+4}$.

$$Y = \frac{2}{(s^2+4)(s^2+9)} = \frac{2/5}{s^2+4} - \frac{2/5}{s^2+9}$$

Answer: $y(t) = \frac{1}{5}\sin(2t) - \frac{2}{15}\sin(3t)$.

Impulse Forcing

Problem. $y'' + 4y = \delta(t-\pi)$, $y(0) = 1, y'(0) = 0$.
Transform: $Y = \frac{s}{s^2+4} + \frac{e^{-\pi s}}{s^2+4}$.
Invert: $\cos(2t)$ for the first term; second shifting for $e^{-\pi s}$:

$$y(t) = \cos(2t) + u_\pi(t) \cdot \frac{1}{2}\sin(2(t-\pi))$$

Since $\sin(2(t-\pi)) = \sin(2t)$, the impulse adds a sine component at $t = \pi$.

The People Behind the Transform

Pierre-Simon Laplace (1749–1827)

French mathematician and astronomer. His *Mécanique Céleste* (5 vols, 1799–1825) developed potential theory and showed the solar system is dynamically stable. When Napoleon reportedly remarked on the absence of God in his celestial mechanics, Laplace is said to have replied that he had no need of that hypothesis. The Laplace transform appears in his *Théorie Analytique des Probabilités* (1812).

Paul Adrien Maurice Dirac (1902–1984)

English theoretical physicist, Nobel Prize 1933 (age 31) for the Dirac equation predicting antimatter. Introduced the delta function in *The Principles of Quantum Mechanics* (1930) as a notational convenience — its rigorous foundation as a distribution came later (Schwartz, 1940s, Fields Medal). The delta function is now indispensable in signal processing, PDEs, and quantum field theory.

The People Behind the Transform (cont.)

Oliver Heaviside (1850–1925)

English self-taught electrical engineer, mathematician, and physicist. Reduced Maxwell's 20 original equations to the 4 now taught. Developed operational calculus — the forerunner of Laplace-transform methods — to solve transmission-line problems. The unit step function bears his name. The etymology is coincidental to the function's being nonzero on one side of the origin.

Niels Henrik Abel (1802–1829)

Norwegian mathematician, dead of tuberculosis at 26. Solved the tautochrone problem at age 18 by what is now called the Laplace-transform method. Proved the general quintic equation has no solution in radicals. Abel's theorem on integrals (1826) — Jacobi reportedly regarded it as the greatest discovery in integral calculus of the nineteenth century. Crelle offered him a Berlin professorship; the letter arrived two days after his death.

One-Page Summary — All the Rules

Definition

$$F(s) = \int_0^\infty e^{-st} f(t)\, dt$$

Exists if f is piecewise continuous + exponential order.

Linearity

$$\mathcal{L}\{\alpha f + \beta g\} = \alpha F + \beta G$$

Shifting Properties

$$\mathcal{L}\{e^{at} f(t)\} = F(s-a)$$

$$\mathcal{L}\{u_a(t) f(t-a)\} = e^{-as} F(s)$$

Derivative Rules

$$\mathcal{L}\{f'\} = sF - f(0)$$

$$\mathcal{L}\{f''\} = s^2 F - sf(0) - f'(0)$$

$$\mathcal{L}\{f^{(n)}\} = s^n F - \sum_{k=0}^{n-1} s^{n-1-k} f^{(k)}(0)$$

Special Functions

$$\mathcal{L}\{u_a(t)\} = \frac{e^{-as}}{s}$$

$$\mathcal{L}\{\delta(t-a)\} = e^{-as}$$

$$\mathcal{L}\{\sin(at)\} = \frac{a}{s^2 + a^2}$$

$$\mathcal{L}\{\cos(at)\} = \frac{s}{s^2 + a^2}$$

One-Page Summary — All the Rules (cont.)

Convolution & Transfer

$$(f * g)(t) = \int_0^t f(\tau)g(t-\tau)\, d\tau$$

$$\mathcal{L}\{f * g\} = F(s)\, G(s)$$

$$H(s) = \frac{X(s)}{F(s)} = \frac{N(s)}{P(s)}$$

$$h(t) = \mathcal{L}^{-1}\{H(s)\}$$

Poles in left half-plane $\Rightarrow$ stable.

The Method

Transform $\to$ substitute ICs $\to$ solve algebraically $\to$ decompose ($F(s)G(s) \to$ convolution; $(s-a)^2 + b^2 \to$ shifting; $(s-r)^k \to$ partial fractions) $\to$ invert.

Three Big Ideas

First: differentiation in t becomes multiplication by s — the entire Laplace method follows from this single substitution.

Second: the convolution theorem is the product rule Laplace transforms always wanted — and it gives Duhamel's principle, transfer functions, and the impulse-response framework for free.

Third: poles in the left half-plane mean stability — and this single geometric picture underpins all of classical control theory.